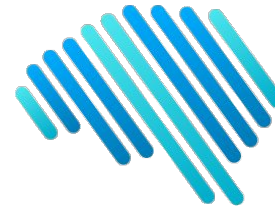




PONTIFICIA
UNIVERSIDAD
CATÓLICA
DE CHILE



IA Lab

Question Answering Models

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Diplomado Inteligencia Artificial

Pontificia Universidad Católica de Chile

Content

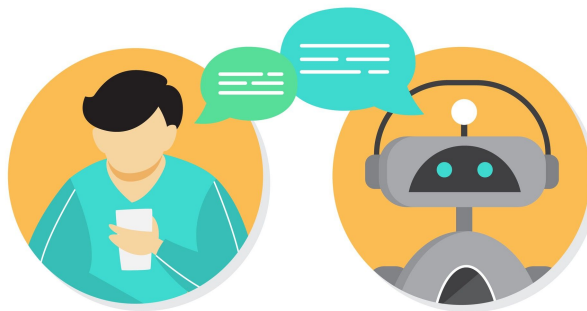
1. Introduction
2. Motivation
3. Areas in QA
 - a. Machine Reading Comprehension
4. Deep Learning Approaches
 - a. Datasets
 - b. Stanford Attentive Reader
 - c. BiDAF
 - d. Transformer-based
5. Metrics
6. References

What is Question Answering?



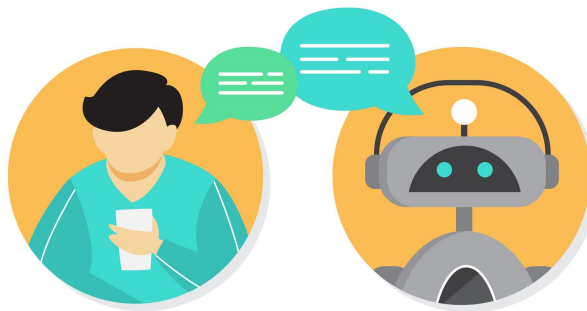
What is Question Answering?

- What time is it?
- What is the weather outside?
- How many did Satchel Paige strike out last night?
- What are those guys doing across the street?



What is Question Answering?

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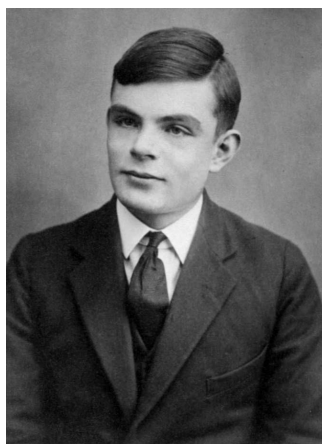


Systems that answer questions posed by humans in natural language query

What is Question Answering?

One of the oldest NLP tasks

- 1950s: “Imitation Game” widely known as “Turing Test”



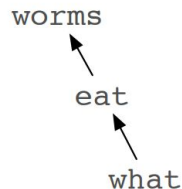
What is Question Answering?

One of the oldest NLP tasks

- 1960s: Punched card systems
 - Indexing and Dependency Logic for Answering English Questions (Simmons et al., 1961)

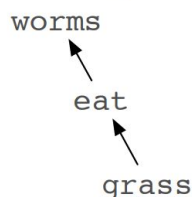
Question:

What do worms eat?

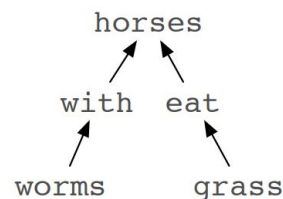


Potential Answers:

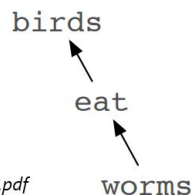
Worms eat grass



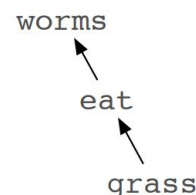
Horses with worms eat grass



Birds eat worms



Grass is eaten by worms



Borrowed from

<https://web.stanford.edu/class/cs124/lec/watsonqa.pdf>

- NLIDB (Androutsopoulos et al., 1993), BASEBALL (Green et al., 1961), LUNAR (Woods et al., 1973), etc

Motivation

Why QA is important?

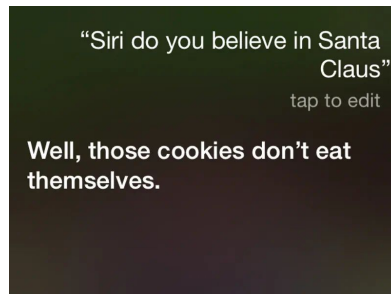
- QA is a complete AI-problem
 - Solving QA also means solving other problems
- Implies many applications
 - Search, IR, dialogue, reading comprehension, etc



Motivation

Why QA is important?

- QA is a complete AI-problem
 - Solving QA also means solving other problems
- Implies many applications
 - Search, IR, dialogue, reading comprehension, etc
- Useful results
 - Siri, Google search, Alexa, etc.
- Much to solve yet
 - Some problems are still unsolved



Let's see Questions

How different are questions?

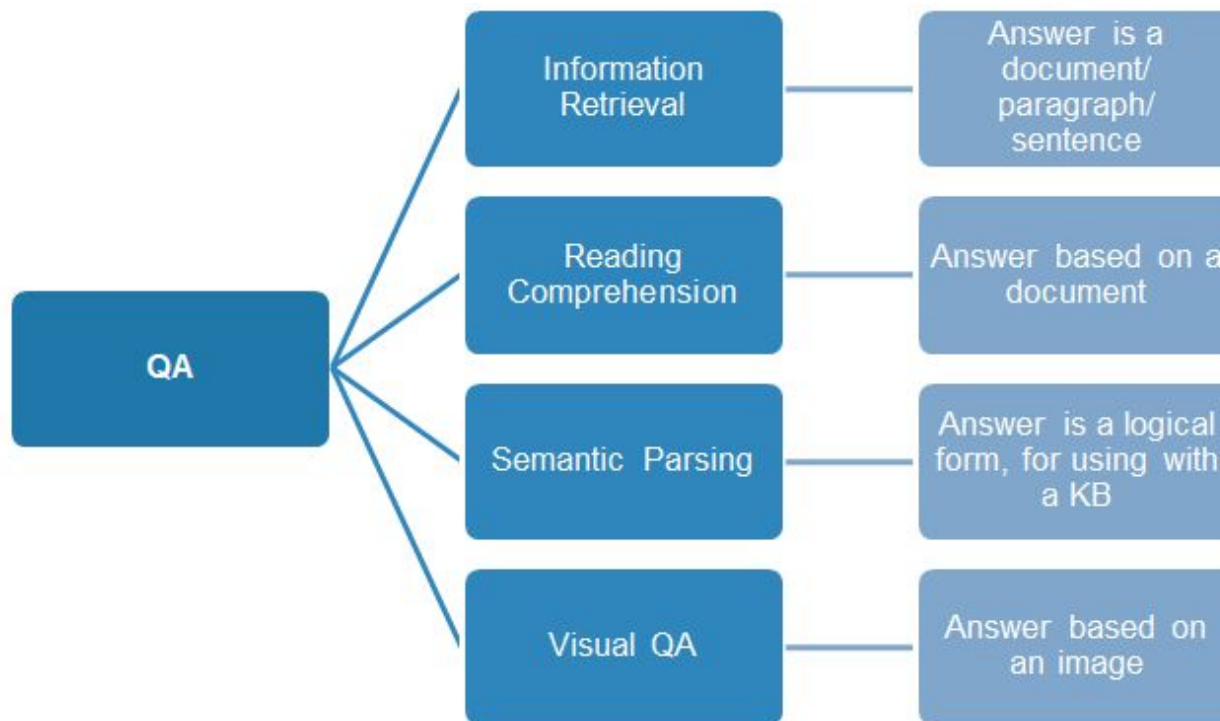
Question	Answer Source
What is a cell?	Encyclopedia, Wikipedia
Where was Barack Obama born?	Biography, Wikipedia
What is the price of an iphone?	Website, Amazon
How is your mother?	Previous knowledge (Memory)
How was the movie?	Personal opinion, observation

Let's see Questions

QA depends on three kinds of data:

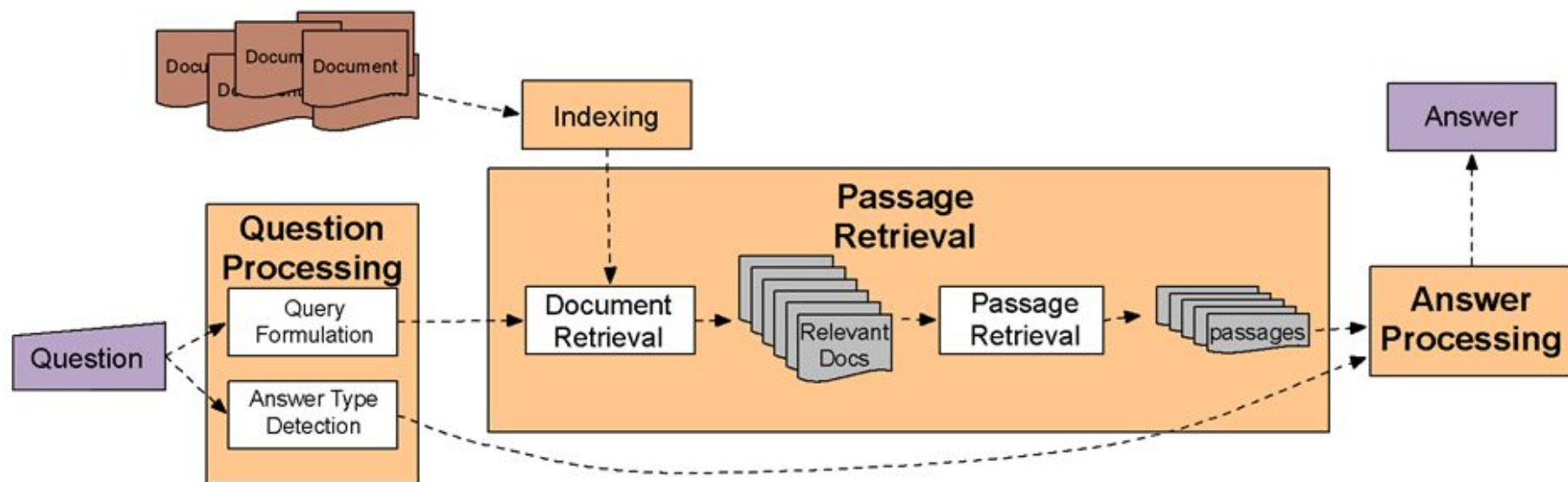
- Question
 - Factual/factoid questions
 - Complex/narrative questions
 - Information retrieval
- Context/Source
 - Corpus, Corpora
 - Knowledge base
 - Non-linguistic sources (sensors, images, etc)
- Answer
 - A single fact
 - A sentence or paragraph (generated or extracted)
 - A document
 - Another question
 - An object (e.g. images)

Areas in QA



We will cover Reading Comprehension in more depth

IR-based Factoid QA

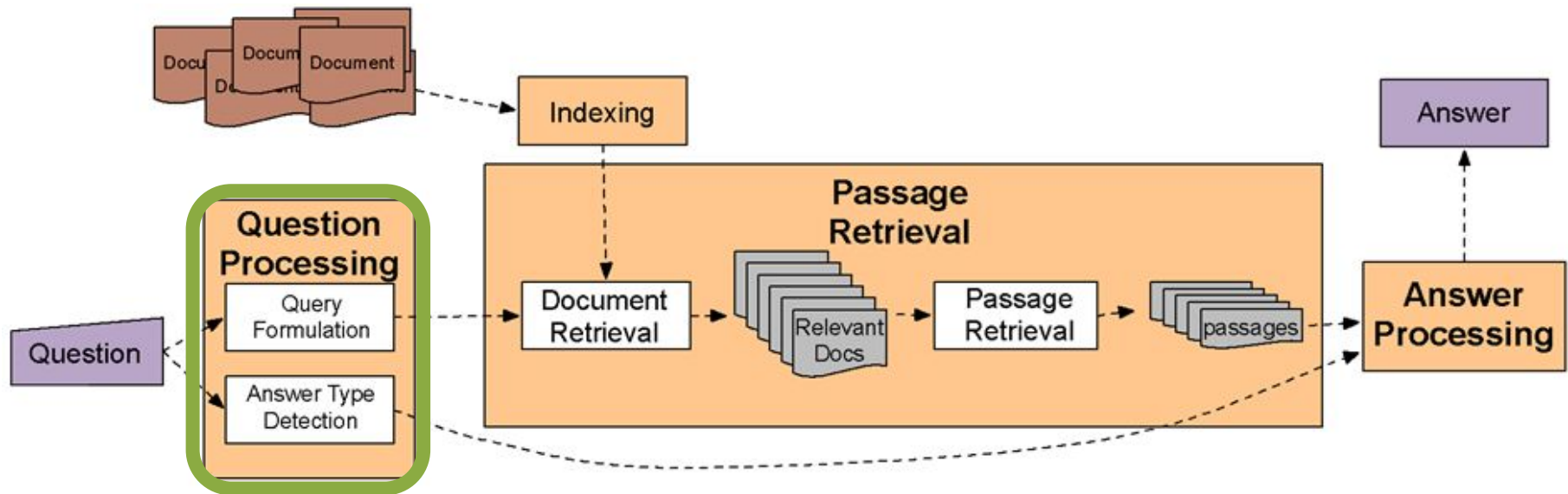


The goal is to answer by finding short text segments on the web or some other collection of documents (Jurafsky & Martin, 2019).

Three principal parts:

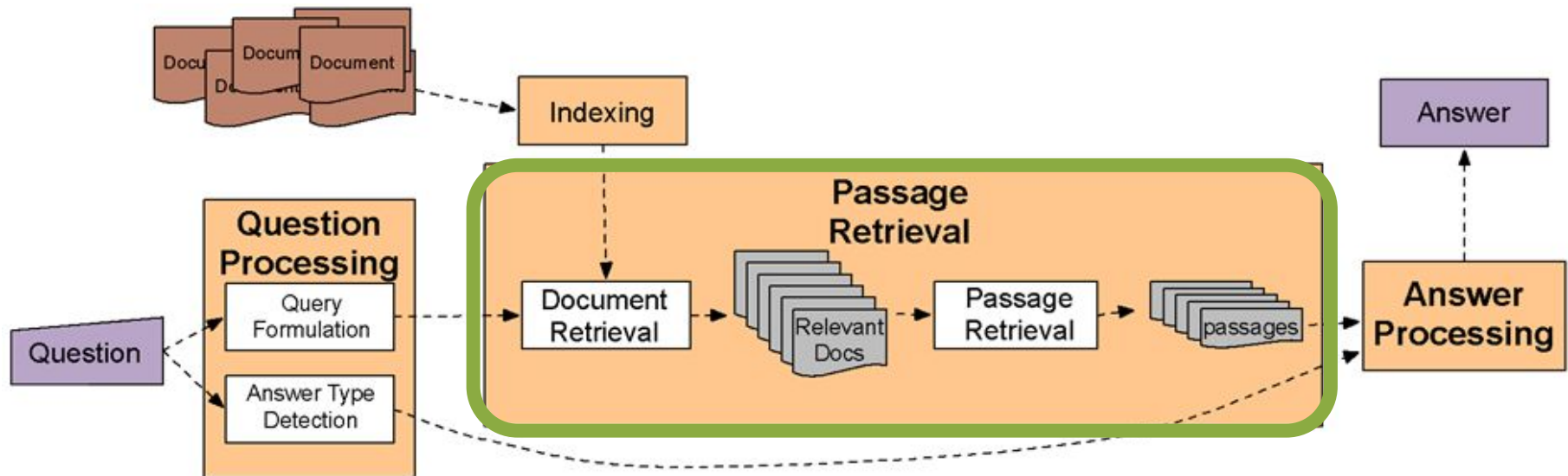
- Question processing
- Passage retrieval
- Answer processing

IR-based Factoid QA



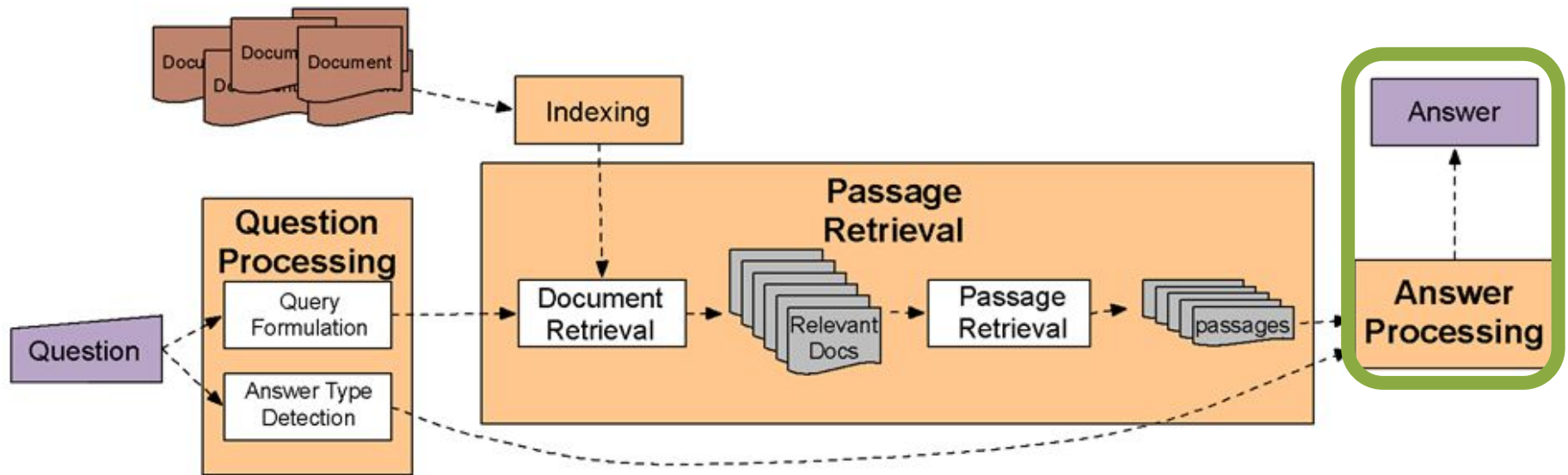
- Question processing
 - Detect question and answer type
 - Detect relations
 - Formulate queries to send to the search engine

IR-based Factoid QA



- Passage Retrieval
 - Retrieve ranked documents
 - Select suitable passages and rerank

IR-based Factoid QA



- Answer processing
 - Extract candidate answers
 - Rank candidates
 - Retrieve N candidates

Reading Comprehension

- Also known as machine reading comprehension (MRC)
- MRC as an instance of QA because it is essentially a question answering problem over a short passage of text
- MRC puts more emphasis on text understanding with answering questions regarded as a way to measure language understanding

Reading Comprehension

Passage (P) + Question (Q) → Answer (A)

P

Alyssa got to the beach after a long trip. She's from Charlotte. She traveled from Atlanta. She's now in Miami. She went to Miami to visit some friends. But she wanted some time to herself at the beach, so she went there first. After going swimming and laying out, she went to her friend Ellen's house. Ellen greeted Alyssa and they both had some lemonade to drink. Alyssa called her friends Kristin and Rachel to meet at Ellen's house.....

Q

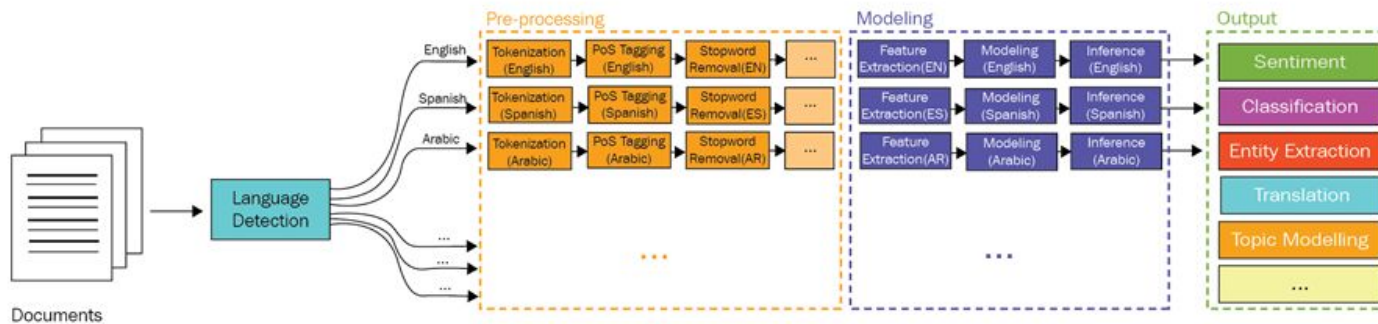
Why did Alyssa go to Miami?

A

To visit some friends

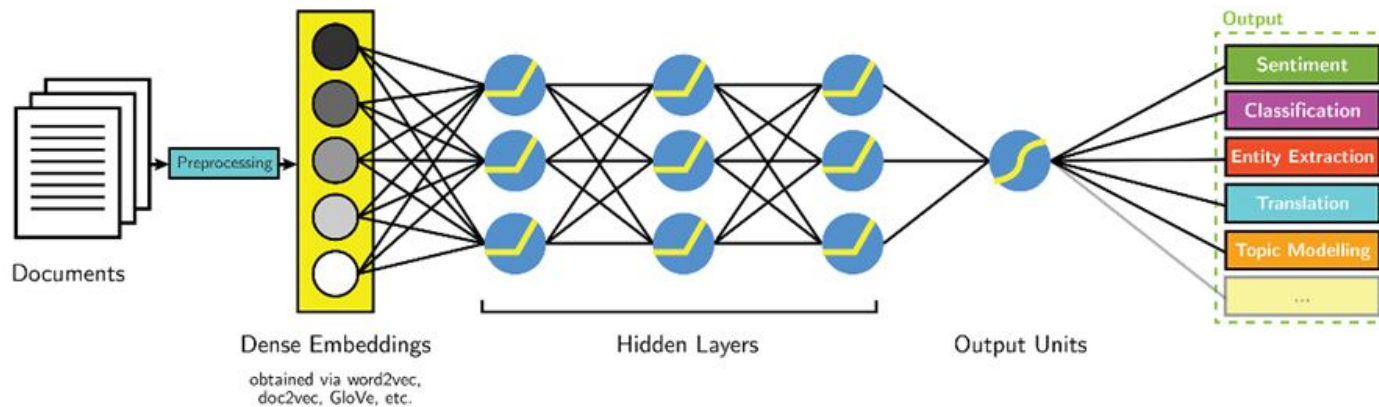
Deep Learning For QA

Classical NLP



Hand-crafted features

Deep Learning-based NLP



Feature extractors in an automatic way (distributed representations)

Generic Neural Network for MRC

Given context c and question q , the probability of the answer a can be represented as:

$$p(a \mid c, q) = \exp(W(a) g(c, q)) \quad \text{where } a \in V$$

- Encode c and q with recurrent networks
- Combine c and q with MLP or attention
- Produce answer a with classifier, attention or generative setup

We Need Data for our Models

Several dataset appears before 2015:



CNN/Daily Mail Datasets



bAbI Children's Book Test



WikiReading



SQuAD



LAMBADA



MS MARCO

CNN Dataset

- Proposed as a *Cloze* task (Hermann et al., 2015)
- Reduces vocabulary by anonymize entities

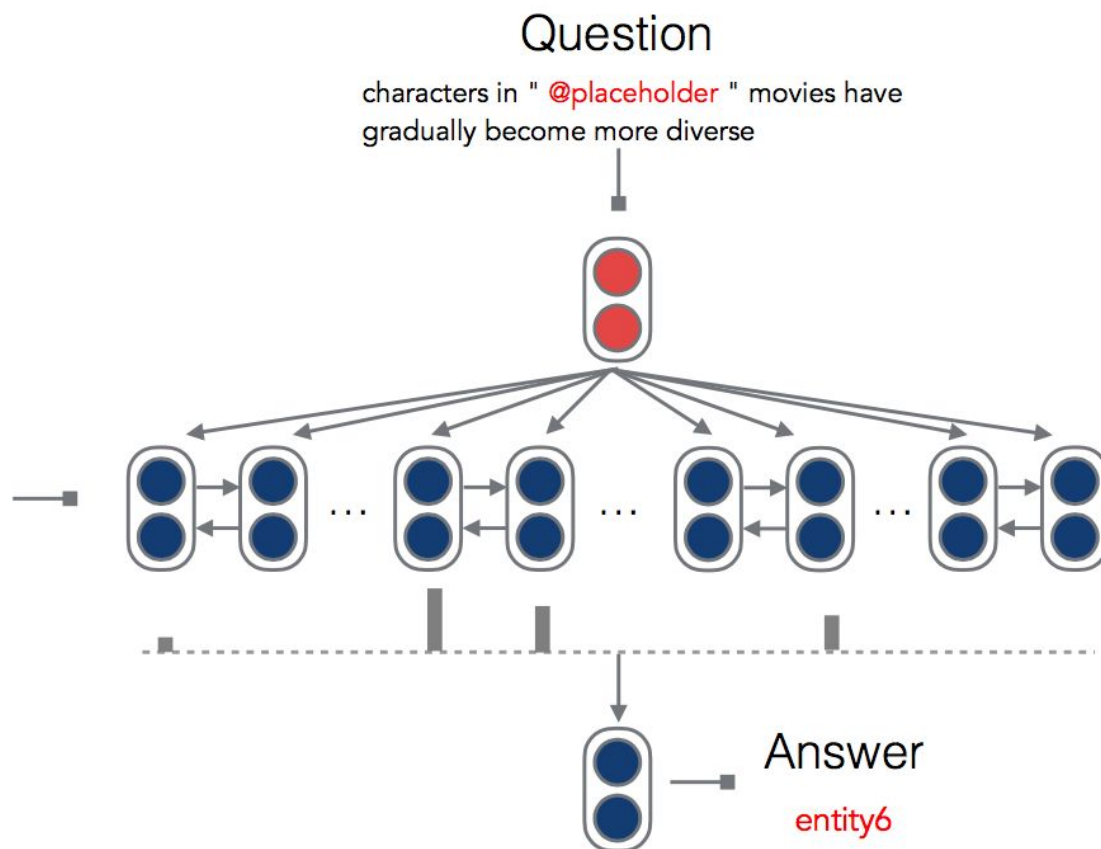
Original Version	Anonymised Version
Context The BBC producer allegedly struck by Jeremy Clarkson will not press charges against the “Top Gear” host, his lawyer said Friday. Clarkson, who hosted one of the most-watched television shows in the world, was dropped by the BBC Wednesday after an internal investigation by the British broadcaster found he had subjected producer Oisin Tymon “to an unprovoked physical and verbal attack.” ...	the <i>ent381</i> producer allegedly struck by <i>ent212</i> will not press charges against the “ <i>ent153</i> ” host , his lawyer said friday . <i>ent212</i> , who hosted one of the most - watched television shows in the world , was dropped by the <i>ent381</i> wednesday after an internal investigation by the <i>ent180</i> broadcaster found he had subjected producer <i>ent193</i> “ to an unprovoked physical and verbal attack . ” ...
Query Producer X will not press charges against Jeremy Clarkson, his lawyer says.	producer X will not press charges against <i>ent212</i> , his lawyer says .
Answer Oisin Tymon	<i>ent193</i>

Stanford Attentive Reader

Demonstrated a minimal, highly successful architecture for reading comprehension and question answering (Chen et. al, 2016)

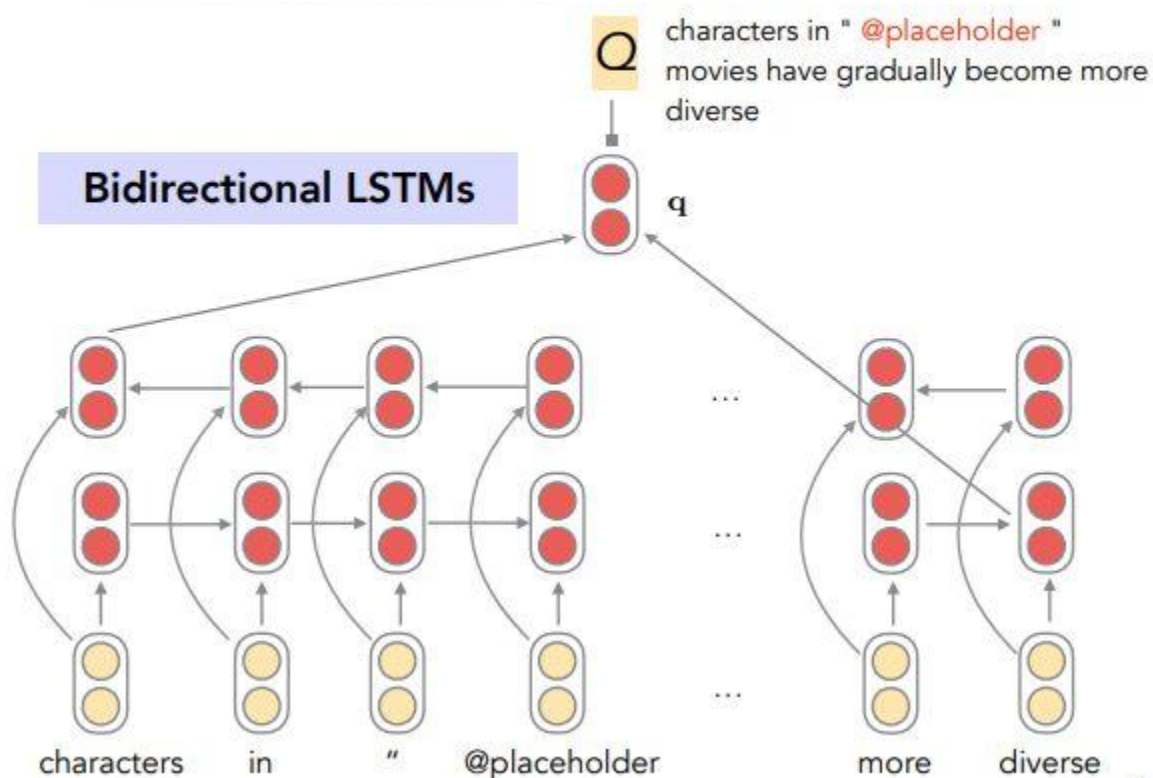
Passage

(@entity4) if you feel a ripple in the force today , it may be the news that the official @entity6 is getting its first gay character . according to the sci-fi website @entity9 , the upcoming novel " @entity11 " will feature a capable but flawed @entity13 official named @entity14 who " also happens to be a lesbian . " the character is the first gay figure in the official @entity6 -- the movies , television shows , comics and books approved by @entity6 franchise owner @entity22 -- according to @entity24 , editor of " @entity6 " books at @entity28 imprint @entity26 .



Stanford Attentive Reader

Uses Bi-LSTM to obtain Q representation



Stanford Attentive Reader

Uses Bi-LSTM to obtain P representation

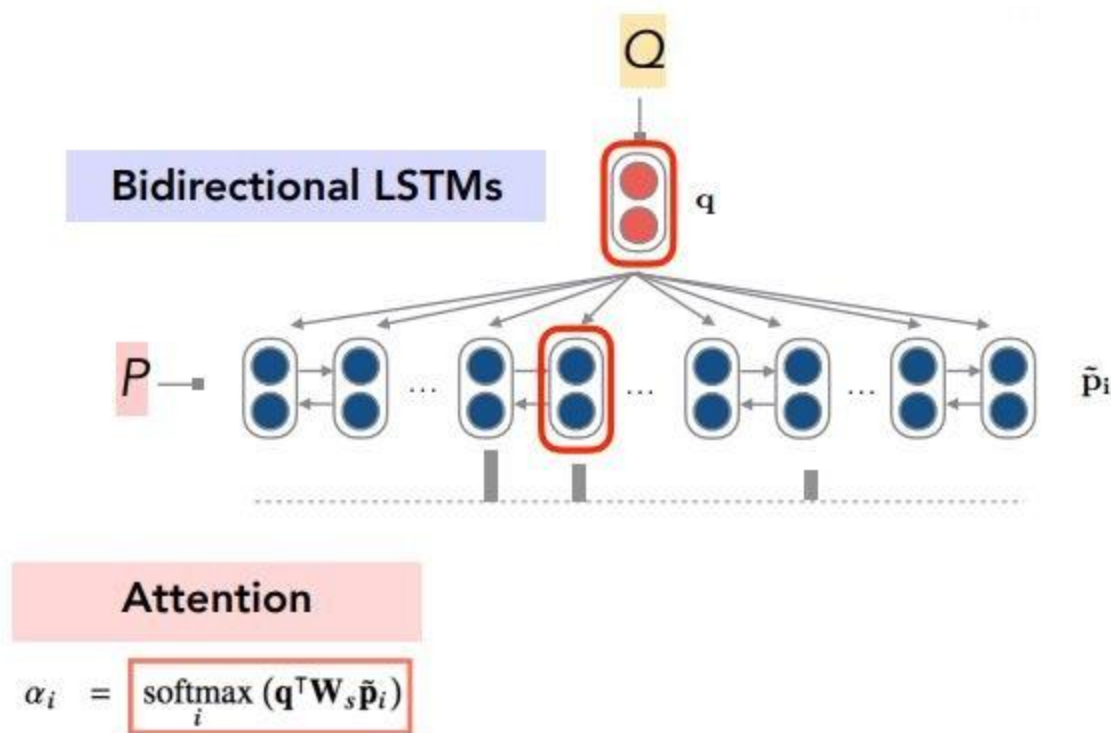
Bidirectional LSTMs



(@entity4) If you feel a ripple in the force today , it may be the news that the official @entity6 is getting its first gay character . according to the sci-fi website @entity9 , the upcoming novel " @entity11 " will feature a capable but flawed @entity13 official named @entity14 who " also happens to be a lesbian . " the character is the first gay figure in the official @entity6 -- the movies , television shows , comics and books approved by @entity6 franchise owner @entity22 -- according to @entity24 , editor of " @entity6 "

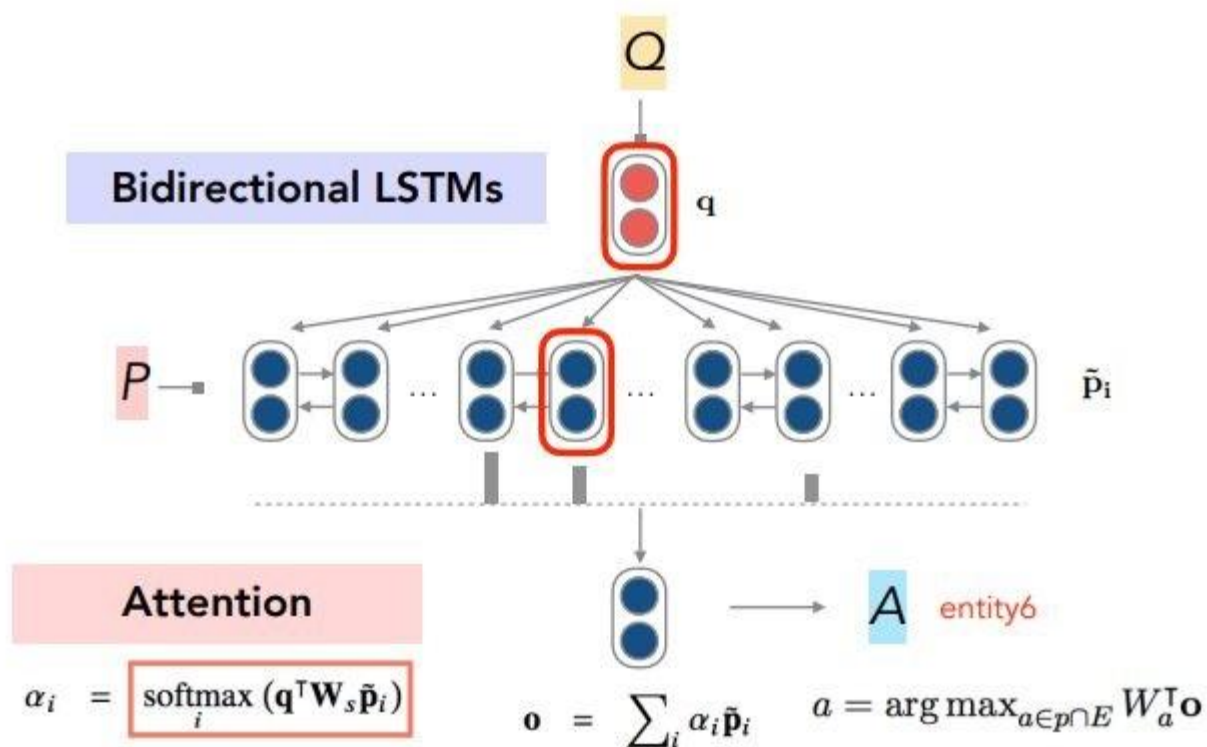
Stanford Attentive Reader

Attention indicates the important token in the passage



Stanford Attentive Reader

Answer **A** predicts the entity



SQuAD Dataset

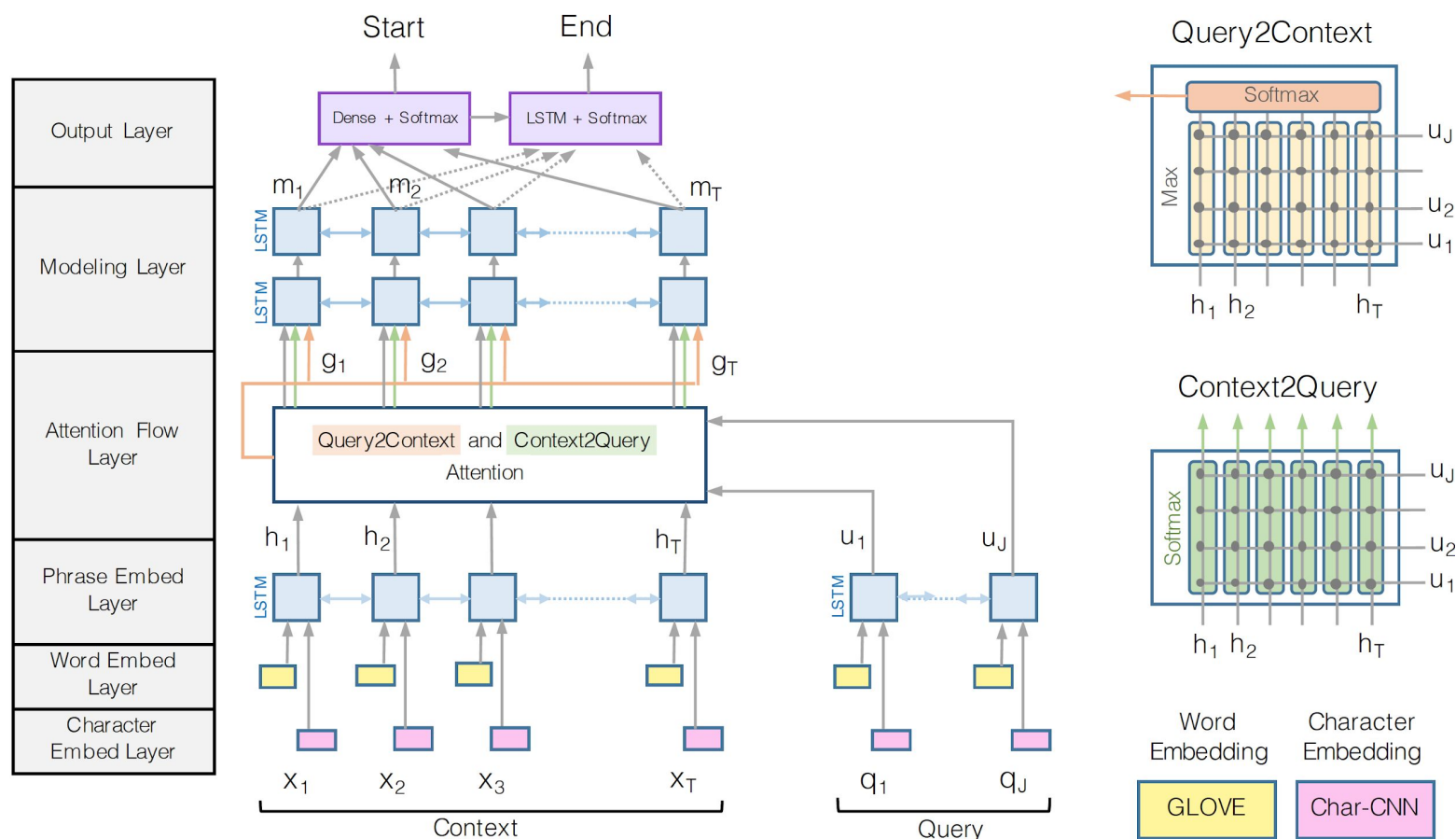
- Task as extractive QA (Rajpurkar et. al, 2016)
- Answer is a span in the passage

The first recorded travels by Europeans to China and back date from this time. The most famous traveler of the period was the Venetian Marco Polo, whose account of his trip to "Cambaluc," the capital of the Great Khan, and of life there astounded the people of Europe. The account of his travels, *Il milione* (or, *The Million*, known in English as the *Travels of Marco Polo*), appeared about the year 1299. Some argue over the accuracy of Marco Polo's accounts due to the lack of mentioning the Great Wall of China, tea houses, which would have been a prominent sight since Europeans had yet to adopt a tea culture, as well the practice of foot binding by the women in capital of the Great Khan. Some suggest that Marco Polo acquired much of his knowledge **through contact with Persian traders** since many of the places he named were in Persian.

How did some suspect that Polo learned about China instead of by actually visiting it?

Answer: **through contact with Persian traders**

BiDAF: Bi-Directional Attention Flow



Idea: attention should flow both ways – from the context to the question and from the question to the context (Seo et. al, 2017)

BiDAF: Bi-Directional Attention Flow

Features:

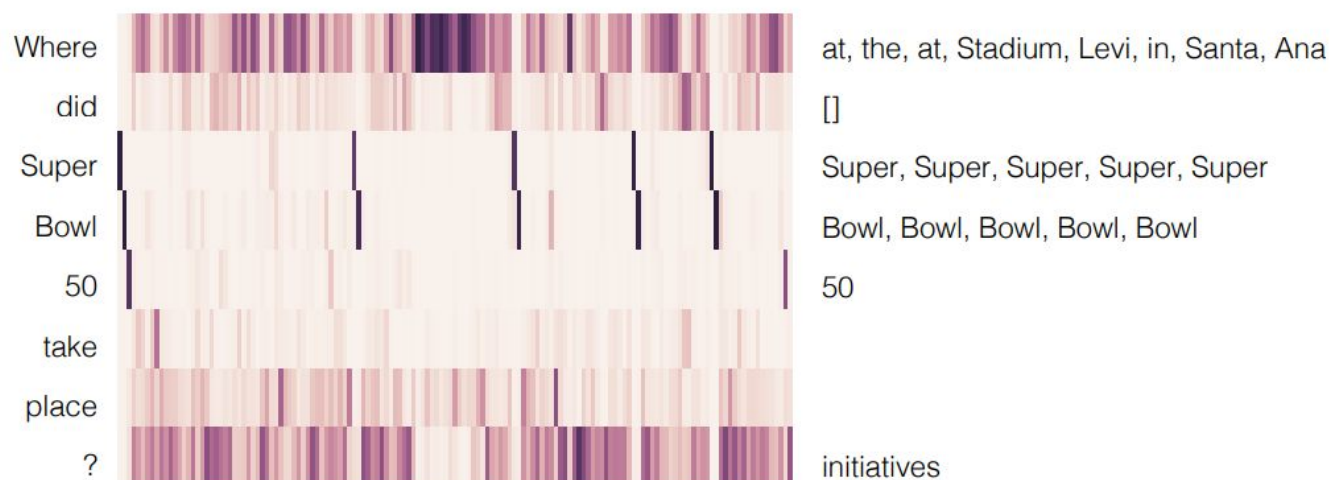
- Pre-trained word embeddings
- Character embeddings
- BiLSTM for contextualization
- Special attention mechanisms:
 - Attention on both question and context
 - Independent for each token

[Dissecting BiDAF](#)

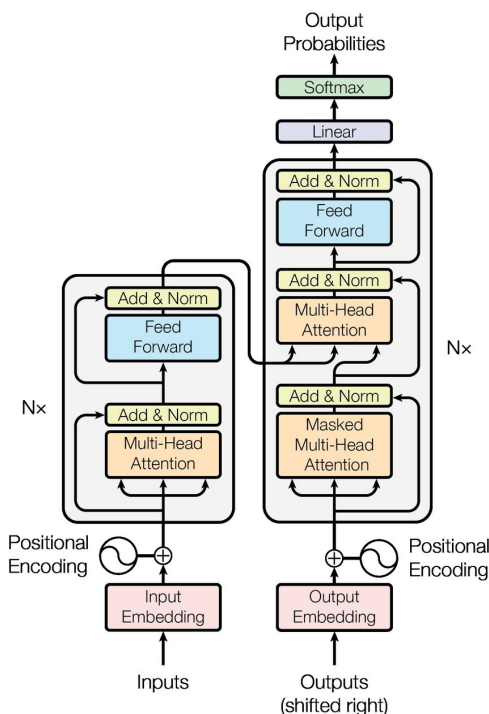
BiDAF: Bi-Directional Attention Flow

We can use the attention weights to find out what information was crucial in giving the answer

Super Bowl 50 was an American football game to determine the champion of the National Football League (NFL) for the 2015 season. The American Football Conference (AFC) champion Denver Broncos defeated the National Football Conference (NFC) champion Carolina Panthers 24–10 to earn their third Super Bowl title. The game was played on February 7, 2016, at Levi's Stadium in the San Francisco Bay Area at Santa Clara, California. As this was the 50th Super Bowl, the league emphasized the "golden anniversary" with various gold-themed initiatives, as well as temporarily suspending the tradition of naming each Super Bowl game with Roman numerals (under which the game would have been known as "Super Bowl L"), so that the logo could prominently feature the Arabic numerals 50.



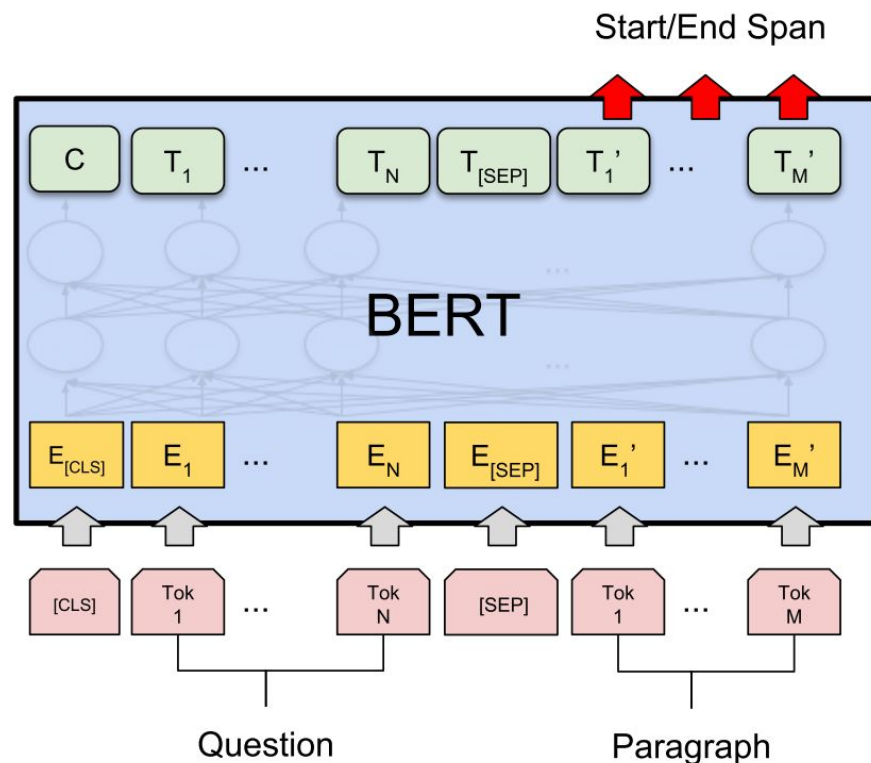
Introduction to the Sesame World



- Transformer-based models (Vaswani et. al, 2017)
- Pre-trained with huge corpora (Devlin et al., 2018)
- Fine-tuned on a variety of tasks (including QA)

BERT for QA

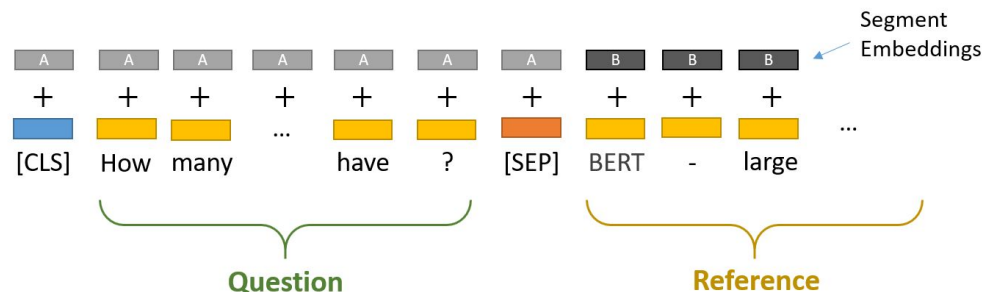
- Take advantage of pre-trained model
- Needs fewer training steps to learn



BERT for QA

Input layer

- Take advantage of segment embedding of BERT
- The question and context are passed to the model together



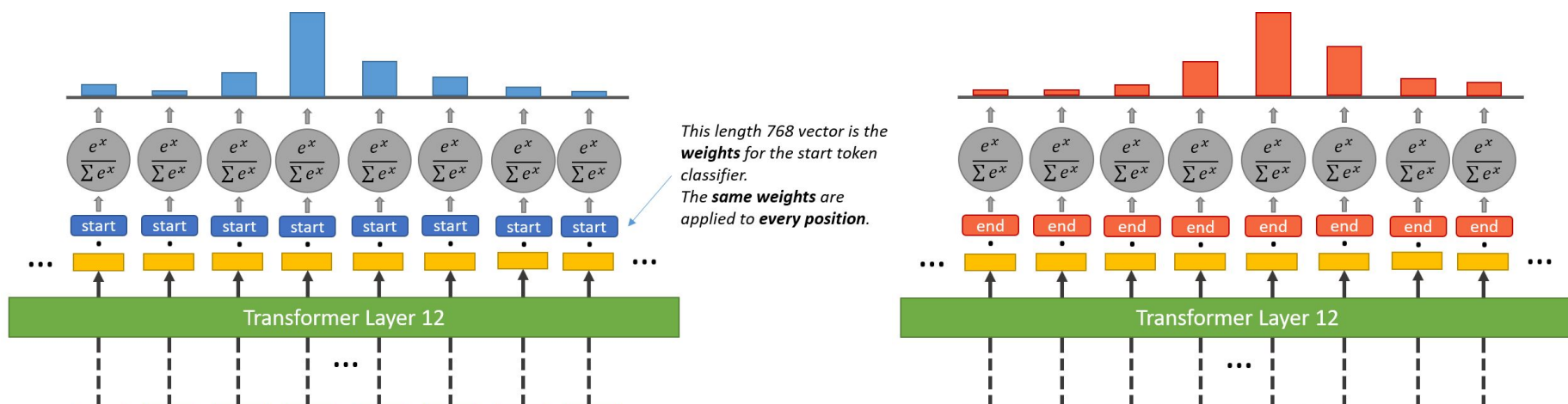
Question: How many parameters does BERT-large have?

Reference Text: BERT-large is really big... it has 24 layers and an embedding size of 1,024, for a total of 340M parameters! Altogether it is 1.34GB, so expect it to take a couple minutes to download to your Colab instance.

BERT for QA

Output layer

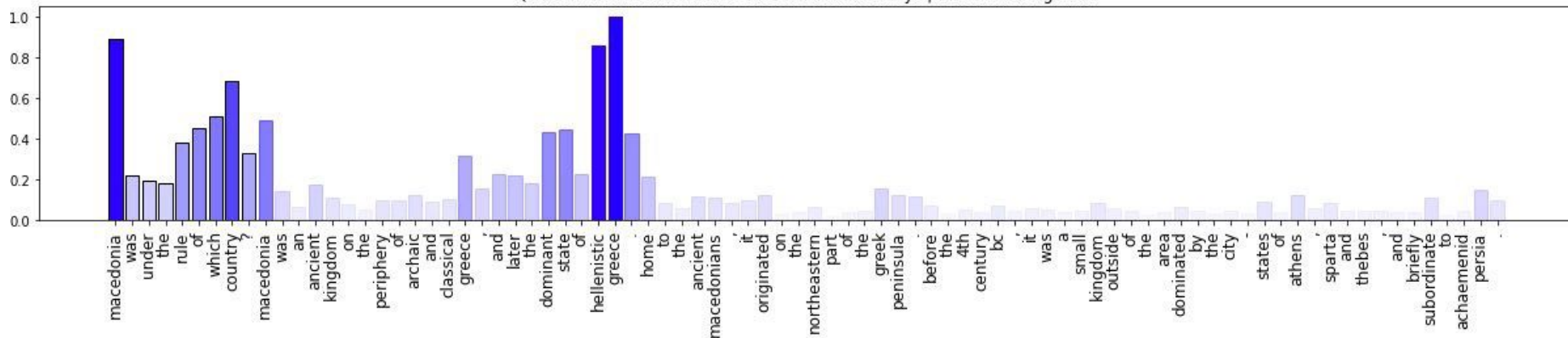
- Two predictors
- Predict start and end span
- Apply prediction layer over every token (position)



BERT for QA

We can use the attention weights to find out what information was crucial in giving the answer

Q: Macedonia was under the rule of which country? | A: hellenistic greece



QUESTION = Macedonia under the rule of which country?

CONTEXT = Macedonia was an ancient kingdom on the periphery of Archaic and Classical Greece, and later the dominant state of Hellenistic Greece. Home to the ancient Macedonians, it originated on the northeastern part of the Greek peninsula. Before the 4th century BC, it was a small kingdom outside of the area dominated by the city-states of Athens, Sparta and Thebes, and **briefly subordinate to Achaemenid Persia.**

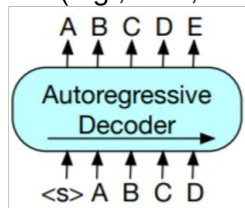
BERT ANSWER: **hellenistic greece**

Generative QA

Transformers (Encoder and Decoder) has been used to create various language models.

Model & illustration

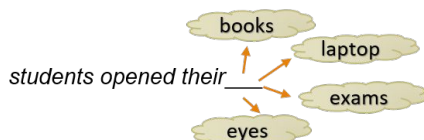
Autoregressive language model (e.g., GPT, GPT-2/3)



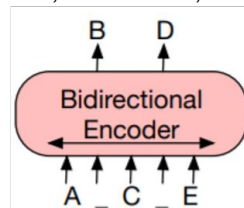
Training objective

Predicting what word comes next given previous words

Example



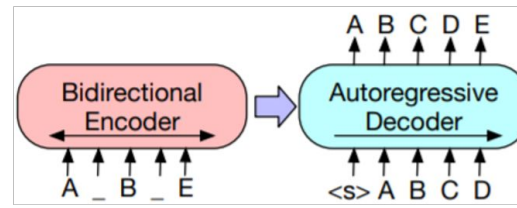
Masked language model (e.g., BERT, RoBERTa, XLM-R)



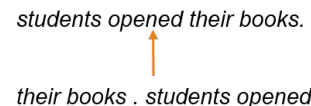
Predicting masked words given other words in the sequence



Encoder-Decoder (e.g., BART, T5)



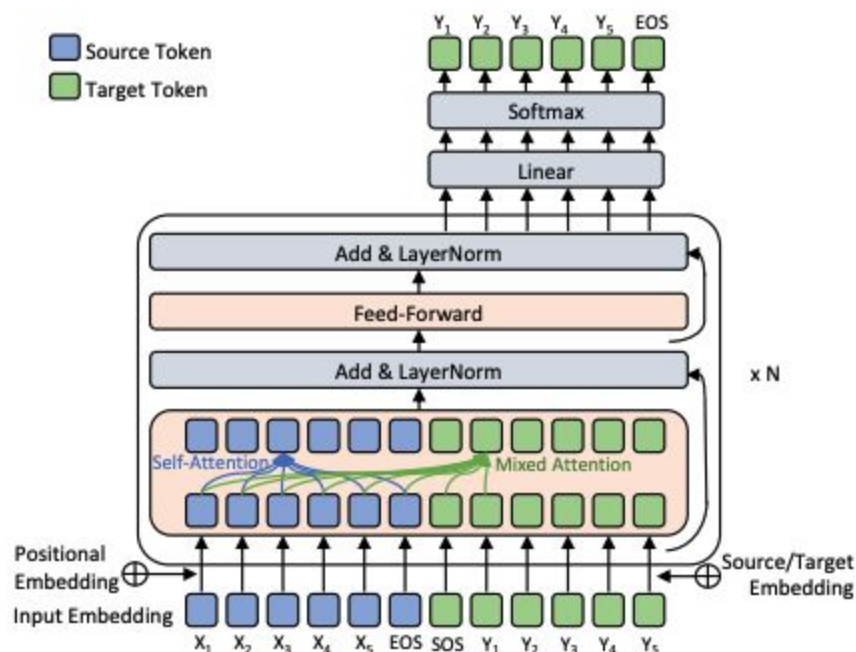
Corrupting a sequence and then predicting the original sequence



Generative QA

GPT fine tuned to generate answers (Radford et al., 2018).

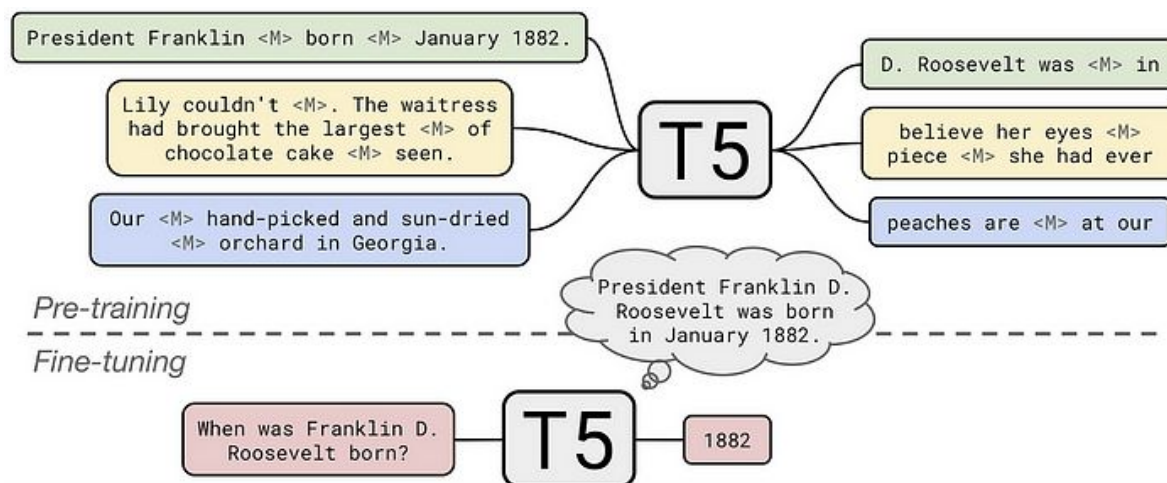
- Provide question + context as the source tokens.
- Target tokens are the generation of each token's answer.



Generative QA

Encoder-Decoder architectures were proposed

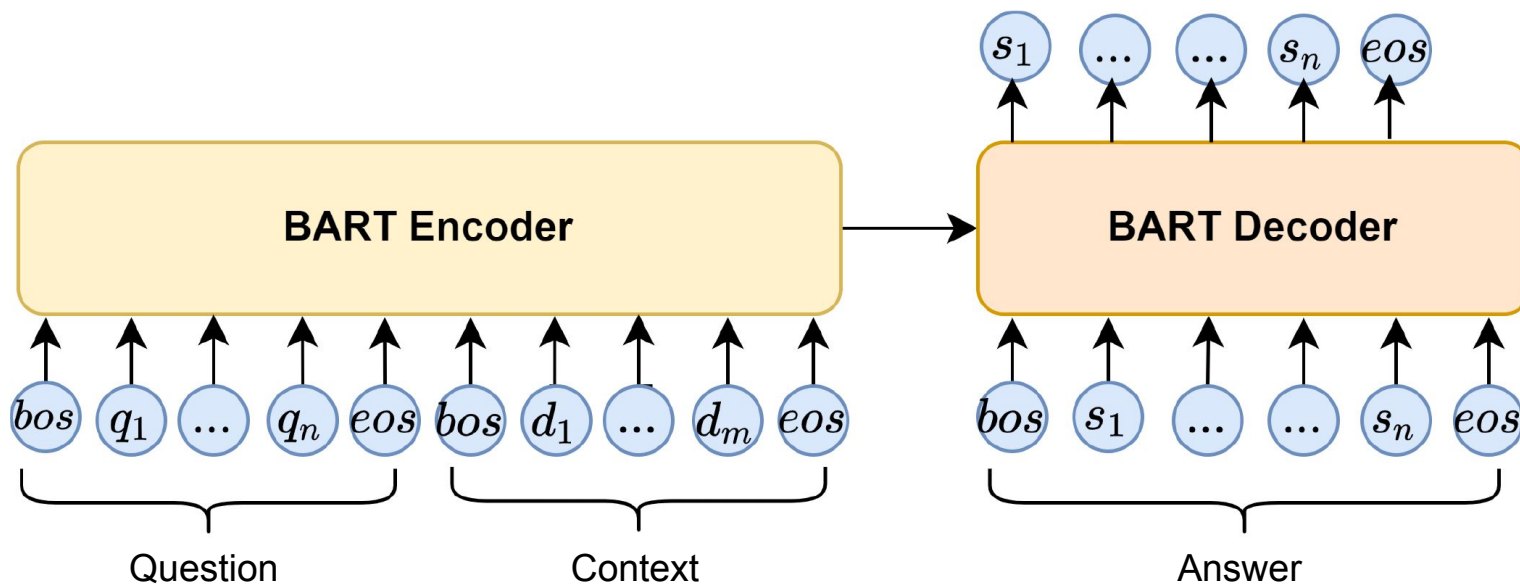
- Pretrained as an autoencoder model to reconstruct the input (e.g., BART) (Lewis et al., 2019)
- Pretrained on a mixture of tasks (e.g., T5) (Raffel et al., 2019)



Generative QA

We can fine tune the model for QA:

- Encoder in charge of processing the input (question + context)
- Decoder in charge of generating the output (answer)



Metrics

Accuracy

- Does answer match gold target?
- Binary measure

$$accuracy = \frac{correct}{correct + incorrect}$$

Mean Reciprocal Rank

- Measures position of first relevant item in return set
 - if first answer is correct: 1
 - else if second: ½
 - Score 0 if answer is not correct

$$MRR = \frac{\sum_{i=1}^N \frac{1}{rank_i}}{N}$$

Metrics

Exact Match

- 1 or 0 accuracy on whether you match one of the 3 answers

F1-Score

- Can be interpreted as a weighted average of the precision and recall
- F1 score reaches its best value at 1 and worst at 0

$$F_1 = 2 * \frac{\text{precision} * \text{recall}}{\text{precision} + \text{recall}}$$

BLUE Score

- Measure for translation accuracy (generation of text)

References

Colaboratory Notebook Tutorial

- <https://github.com/vgaraujov/Question-Answering-Tutorial>

Papers

- Daniel Jurafsky & James H. Martin, Speech and Language Processing, 2019.
- Bolanle Ojokoh¹ and Emmanuel Adebisi, A Review of Question Answering Systems, 2019.

Slides

- Christopher Manning, Natural Language Processing with Deep Learning.
- Karl Moritz Hermann, Deep Learning for Natural Language processing: Question Answering
- Deepak Gupta, Question Answering: Learning to Answer from Text